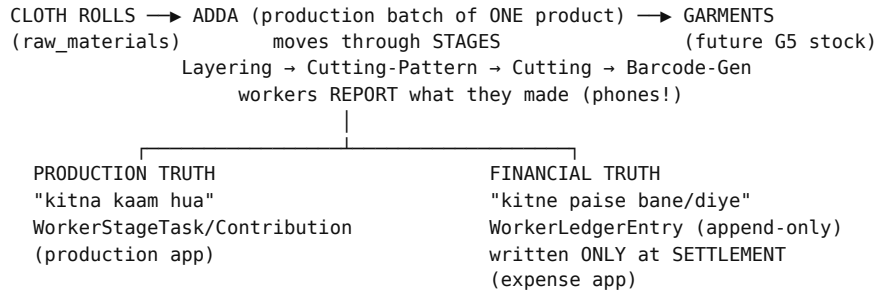


PROJECT KNOWLEDGE MAP — Kapil Enterprises ERP

Mandatory first read for every new developer — and for the owner learning the system. Business flow → architecture → database → code. Companion reading order: [LEARNING PATH.md](#) · every active doc: [DOCUMENTATION INDEX.md](#). Stack: Django 5.0.1 + PostgreSQL, server-rendered, mobile-first (rule 11).

1) The business in one diagram



One sentence: **workers report work** → **owner settles the Adda (earnings book)** → **owner pays cash later** → **everything is traceable forever**.

2) Request flow (what happens on one click)

```
Browser → urls.py → SidebarAccessMiddleware (menu hidden ⇒ URL blocked)
→ View (permission_service / skill+assignment gates;
    | NEVER writes more than one row itself)
→ SERVICE (config/<app>/services/ – ALL multi-row writes,
    | @transaction.atomic, explicit – NO signals, ADR-0001)
→ Models / PostgreSQL (constraints enforce truth even against bugs)
→ history_service (timeline event, append-only)
← redirect + message → server-rendered template (base.html canon CSS)
```

3) Production-truth flow (the work story)

```
manager: set_stage_workers(sr, [ids]) ← SOLE WST writer (CI gate 4/4)
    ↳ WorkerStageTask rows (≤1 ACTIVE per (stage_record, worker);
        un-tick = CANCELLED, never deleted)
worker phone: report_contributions(task, lines)
    ↳ WorkerStageContribution rows (reported_quantity IMMUTABLE – owner $6)
worker: complete_worker_task
    ↳ FREEZES expected_rate / expected_earning (visibility, NEVER money)
management (P1): set_verified_quantity ← correction, reported untouched
stage completes: unreported tasks AUTO-CANCEL (F3) – P2 dialog names them first
```

C-TM (locked): every capture path — manual today, barcode scans tomorrow — converges into WSC through `worker_task_service`. No second door, ever.

4) Financial-truth flow + settlement lifecycle

```

    (reads production truth; NEVER edits it)
settlement queue → DRAFT (ADST-0007; scratchpad, no money)
  | discard ok
  ▼ FINALIZE — adda_settlement_service, lock order:
  |   advisory 5374 → ADST row → stage records → profiles → advances
  |— per line: SWA earning line + ledger CREDIT (stage_earning)
  |   qty = verified-else-reported × frozen expected_rate
  |— per chosen advance: ledger DEBIT (advance_recovery) + PSI row
  |— per worker: FROZEN AddaSettlementItem (what the owner approved)
  |— AddaHistory: SETTLEMENT_FINALIZED
  ▼ mistake?
REVERSE (or REVERSE+SUPERSEDE → successor draft, chain kept)
  |— compensating ledger rows · SWA voided · PSI stamped reversed_at
  ▼ cash day (separate event!)
PayrollSettlement (payment-ONLY) → ledger DEBIT (settlement_payment)

```

Worker sees three NON-overlapping numbers: **Expected (unsettled)** → **Earned (settled)** → **Paid (cash)** — proven by tests.

Eras (ADR-0007, cutover EXECUTED in V2-3): old rows credited at allocation = era-A (`SWA.adda_settlement` NULL, readable+reversible forever); new world = settlement-first (`LEDGER_CREDIT_AT_ALLOCATION` default False; env True = tested rollback lever; physical deletion soak-gated).

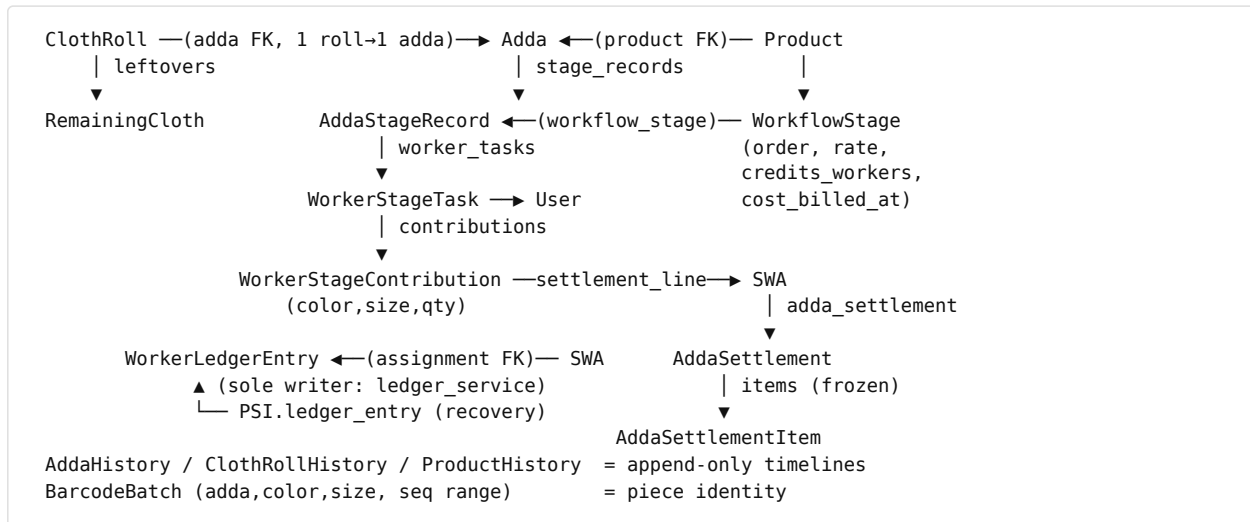
5) Adda lifecycle

```

START (one click: pick product → code auto e.g. 3-PATTI-003)
  ↳ AddaStageRecord per WorkflowStage (per-product flow, owner-configured
    in the Flow Editor: order, rate, billed-at grouping, pay-eligibility,
    future Tracking Mode TM-1)
LAYERING rolls attach (weight verified) ... leftovers MANDATORY at complete
CUTTING-PATTERN evidence (photos/video)
CUTTING breakup per size, bundles, workers report color/size/qty ← first
  real quantities; duration auto from timestamps (owner rule)
BARCODE-GEN BarcodeBatch ranges per (adda,color,size) — piece identity =
  (adda, seq), printed payloads PERMANENT (ADR-0010 §3)
COMPLETE → settle → pay. REOPEN allowed until money: settlement-credited
  stage refuses reopen, names the ADST (V2-3 armor).

```

6) Database relationships (core ER sketch)



FK policy: PROTECT on every money/history anchor (a financial row's parents can never vanish) · CASCADE only where a child is meaningless without its parent · SET_NULL for optional references.

Full service-connection diagram: [LEARNING/04_SERVICE_LAYER.md](#).

7) The five chokepoint services (why they exist, what breaks without them)

Service	Sole writer of	Who may call	Invariant it protects	If removed/bypassed
<code>worker_task_service</code>	WST + WSC (+expected freeze, verified qty)	views/stage services; workers only via report view	roster = participation; reported immutable; ALL capture paths converge (C-TM)	second truth source → settlement pays unverifiable numbers
<code>adda_settlement_service</code>	AddaSettlement(+Item), era-B SWAs	management UI only	a line is paid AT MOST ONCE; full provenance item ↔ SWA ↔ ledger ↔ WSC; strict lock order	double-pay, deadlocks, unauditable money
<code>ledger_service</code>	WorkerLedgerEntry	other expense services only	append-only money; reversal nets in-period	silent balance corruption, permanent
<code>allocation_service</code>	era-A SWAs (+era-A void)	lever-ON only (rollback)	symmetric double-credit guard; era-B lines refuse void	legacy path racing settlements
<code>cost_service</code>	processing_cost freeze; <code>role_rate_for</code>	stage services	standard-cost frozen price-at-time; grouped members yield NO rate (C-1 — no double pay)	retro cost rewrites; grouped double-pay

(`history_service` = sixth: every `*History` row. `consume_leftover` = material-consumption writer, C-1.)

8) Phase history (how we got here)

Phase	Shipped	One-line why
V2-1 a → d (2026-06-09 → 11)	WST/WSC truth layer; M2M dropped (0035)	per-worker truth instead of a bare roster
P2 + validation	schema-driven worker report UI, phone-first	workers can actually report
R1	locked Stage Contract + archetypes + scan seam	stages stay open-closed forever
V2-2 PR-A → D (2026-06-11)	AddaSettlement event: finalize/reverse/supersede + mgmt UI	money at the RIGHT moment, corrections first-class
V2-3 (2026-06-11)	settlement-FIRST default + money armor + Expected/Earned/Paid	one default money path; legacy = lever
C-1 (2026-06-11)	grouped-member guard, leftover write path, ADR-0009/0010	regret-prevention before real data
P1/P2/P3 (2026-06-12)	verified-qty surface, completion warning, duality hint	normal human mistakes stop costing pay
Frontend S1 + A-scope (2026-06-12)	mobile table fix, vendor central, canon CSS vocabulary	money screens = reference UI for G4/Missing/Alter

9) ADR summary (locked decisions — contradicting one needs a new ADR)

ADR	Lock
0001	services own all multi-row writes; NO signals
0002	one writer per ledger/history table
0003	RBAC = type/role/skill; stage access skill-gated
0004	tracking = append-only history primitive
0005	production truth ≠ financial truth (Option B; expected_* = visibility)
0006	architect for scale, don't implement early
0007	era coexistence + lever; cutover EXECUTED; deletion soak-gated
0008	commerce boundary: Order ↔ Adda only via stock; revenue never in manufacturing (+margin addendum)
0009	cost duality NEVER additive; full-cost formula; labor-source rule; purchase-price semantics
0010	global references forever; multi-factory = one DB + site dim (never fork); barcode payloads permanent; rework case-scoped; no price fields in production

10) Roadmap (canonical: ROADMAP_REVIEW_POST_C1_2026_06_11.md)

DEPLOY → soak (G4 thin slice rides along; TM-1 week 1-2)
→ MissingPiece → Alter/Rework → G1 material costing → G3 variance ₹
→ [soak gate: ADR-0007 deletion PR]
→ G6 SKU → G5 finished-goods stock → G2 orders/revenue
→ Reporting (full) → G7 planning. TM-2 = Barcode/Traceability review.

11) Standing rules every contributor inherits

Mobile-first is FUNCTIONAL (rule 11): worker flows phone-first; management responsive; every UI review documents 360/768/1280 verification · honest=NULL (unknown ≠ zero) · backfill only known facts · work-that-happened is immutable · soft-state over delete · one writer per truth table (CI gates 4/4, 4b/4, 4c/4) · gate must pass (`bash scripts/check.sh`).